

Preliminary Project Report 3:

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Topic: AI-Assisted Forensic Enhancement and Minutiae Localization in Partial Latent Fingerprints Using Pix2Pix Networks

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Github : <https://github.com/sdShaggy/GAFIS>

Introduction:

Preliminary Project Report 2 established the dataset foundation of this work by performing a statistical, quality-based, and preprocessing analysis of the SOCOFing Real Fingerprint Database, followed by the design of a synthetic latent fingerprint degradation framework (Gaussian blur, contrast reduction, partial occlusion, and information loss simulation). That report confirmed that the SOCOFing dataset is statistically consistent, of high ridge-valley quality, and suitable for controlled degradation experiments, and it established a baseline three-level degradation scheme (mild, moderate, severe).

This report, Preliminary Project Report 3 is the most extensive part of the report and it extends that foundation in two directions. First, it refines the degradation framework by incorporating real sensor noise characteristics drawn from the PolyU Cross-Sensor and High-Resolution Fingerprint datasets, so that the synthetically generated degraded prints more closely approximate genuine forensic latent impressions rather than purely parametric blur and occlusion. Second, it develops the complete theoretical foundation of the Pix2Pix conditional generative adversarial network (cGAN) that will be used to learn the mapping from degraded (latent-like) fingerprints back to their clean, identity-preserving counterparts. This theoretical treatment covers the classical and deep learning enhancement literature, the mathematics of conditional adversarial training, the U-Net generator and PatchGAN discriminator architectures, and the anticipated downstream impact on YOLO-NAS based minutiae localization and SourceAFIS based matching.

The report is intentionally written as a methodology chapter rather than an implementation log, so that it can be reused directly as the theoretical backbone of the dissertation and the eventual research paper once experimental results are available.

Section 1: PolyU-Inspired Realistic Fingerprint Degradation:

1.1 Motivation

Forensic latent fingerprints recovered from crime scenes differ substantially from rolled or plain impressions captured under controlled sensor conditions. Latent prints are typically partial, low in contrast, contaminated by background clutter, and affected by non-uniform pressure and surface texture during deposition. A purely parametric degradation model, of the type used in Preliminary Project Report 2 (Gaussian blur, contrast reduction, rectangular occlusion), captures the geometric symptoms of degradation but does not reproduce the statistical texture of real sensor and background noise. A generator trained only on parametric degradations therefore risks overfitting to synthetic artefacts that do not generalize to genuine forensic samples.

To reduce this synthetic-to-real domain gap, the degradation pipeline is extended using background and noise characteristics sampled from the PolyU fingerprint databases, which include cross-sensor and high-resolution impressions captured under varying pressure, moisture, and contact conditions. The PolyU data is not used as a training target for the generator; the SOCOFing dataset remains the sole source of clean ground-truth identities. Instead, PolyU imagery contributes only its noise texture and background statistics, which are blended onto SOCOFing prints to produce more forensically realistic degraded inputs.

1.2 Limitations of Purely Synthetic Degradation

- Parametric blur kernels produce smooth, spatially uniform degradation that does not match the granular, spatially correlated noise found in real latent lifts.
- Rectangular or elliptical occlusion masks create hard geometric boundaries rarely observed in genuine partial prints, where information loss follows irregular contact regions.
- Sensor-specific artefacts, such as optical sensor smearing, capacitive sensor dropout, and paper or surface texture bleed-through, cannot be reproduced by additive noise alone.
- A generator trained exclusively on parametric degradation may learn to invert the specific noise model rather than genuine ridge restoration, limiting generalization to unseen forensic samples.

1.3 PolyU Dataset Characteristics

The PolyU fingerprint databases used in this work provide cross-sensor and high-resolution impressions collected with optical and different-contact sensors, including variation in illumination, moisture, and applied pressure. Relative to SOCOFing, PolyU samples exhibit lower and less uniform contrast, visible background texture, and localized intensity gradients caused by uneven finger contact. These properties make PolyU imagery a suitable statistical source for background and noise transfer, even though its subject identities are not used anywhere in the paired training set.

Property	SOCOFing (Target Domain)	PolyU (Noise Source)
Acquisition	Controlled optical scanner	Cross-sensor, variable contact
Contrast	High, uniform	Variable, often reduced
Background	Clean, near-uniform	Textured, sensor-specific
Role in pipeline	Ground truth identity	Background / noise donor only

1.4 Proposed Degradation Methodology

The refined degradation pipeline operates in three stages. In the first stage, a bank of PolyU background and noise patches is extracted and characterized in terms of local mean intensity, local variance, and spatial frequency content. In the second stage, a clean SOCOFing fingerprint image is degraded using the parametric operators established in Report 2 (Gaussian blur, contrast attenuation, ridge dropout, elastic distortion, and irregular occlusion). In the third stage, a PolyU-derived noise and background patch is blended onto the parametrically degraded image using alpha compositing in both the spatial and frequency domain, so that the resulting image simultaneously exhibits geometric degradation and realistic sensor-level texture.

SOCOFing Clean Print --> Parametric Degradation --> PolyU Texture / Background Extraction
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v
Frequency-Domain Blend --> Alpha Compositing --> Realistic Synthetic Latent Print

Formally, let I represent the clean SOCOFing fingerprint, $D(I)$ the output of the parametric degradation operators described above, and N a PolyU-derived noise and background patch resized to match I . The composite degraded image I' is generated using a spatially varying blending coefficient $a(x,y)$ that controls the local contribution of the noise patch:

$$I'(x,y) = a(x,y) \cdot D(I)(x,y) + (1 - a(x,y)) \cdot N(x,y)$$

The coefficient $a(x,y)$ is drawn from a smoothed random field so that blending strength varies gradually across the image, avoiding the hard edges produced by simple rectangular occlusion. Regions with lower $a(x,y)$ correspond to areas of greater simulated information loss, analogous to smudging or poor surface contact in real latent prints.

1.5 Expected Benefits

- Reduces the statistical gap between synthetically degraded training pairs and genuine forensic latent impressions.
- Introduces spatially correlated, sensor-realistic noise rather than purely additive or geometric artefacts.
- Improves the generalization of the Pix2Pix generator to fingerprint samples outside the SOCOFing distribution.
- Preserves the identity-preserving pairing required for supervised training, since the ground truth target remains the original, undegraded SOCOFing image.

This refined degradation framework forms the input side of the paired dataset used to train the Pix2Pix enhancement network described in the remainder of this report.

Section 2: Pix2Pix Conditional GAN Theory for Fingerprint Enhancement:

2.1 Motivation for Deep Learning Based Enhancement

Fingerprint enhancement is a necessary precursor to reliable minutiae detection and matching. Degraded ridge structures directly reduce the number and confidence of detected minutiae, which in turn lowers matching accuracy. Classical enhancement techniques rely on hand-crafted assumptions about ridge geometry and noise statistics, which hold reasonably well for rolled or plain impressions but degrade sharply on partial, low-contrast latent prints. Deep learning based enhancement, and specifically image-to-image translation using conditional generative adversarial networks, offers a data-driven alternative that learns the degradation-to-restoration mapping directly from paired examples rather than relying on fixed filters.

2.2 Fingerprint Recognition Pipeline Theory

Automated fingerprint recognition proceeds through a sequence of dependent stages, and degradation at the image level propagates through every subsequent stage:

Ridge / Valley Structure --> Minutiae Extraction --> Descriptor Formation --> Matching

Ridge and valley structure must first be clearly resolved, since it is the raw substrate from which minutiae, the ridge endings and bifurcations used for identification, are extracted. Minutiae are then encoded into descriptors that capture their spatial position, orientation, and local neighbourhood, and these descriptors are finally compared between a probe and a reference print to compute a similarity or matching score. Because each stage depends on the fidelity of the previous one, any loss of ridge information caused by blur, contrast loss, or occlusion is compounded through minutiae extraction and descriptor matching, which is why enhancement is positioned as an upstream, rather than optional, stage of the pipeline.

2.3 Classical Enhancement Approaches and Their Limitations

A range of classical, model-based enhancement techniques have historically been used for fingerprint images, including Contrast Limited Adaptive Histogram Equalization (CLAHE), Gabor filtering, Fourier and Short-Time Fourier Transform (STFT) based frequency filtering, and orientation field estimation followed by directional filtering.

- CLAHE improves local contrast through tile-based histogram equalization but cannot recover ridge information that has been genuinely lost through blur or occlusion; it can only redistribute the contrast that remains.
- Gabor filtering enhances ridges that follow a locally consistent orientation and frequency, but performs poorly in regions of high curvature, ridge discontinuity, or where the underlying orientation field cannot be reliably estimated due to noise.
- FFT and STFT based frequency filtering assumes a locally periodic ridge pattern and can amplify noise in regions where this periodicity assumption breaks down, such as near scars, creases, or occluded areas.
- Orientation field estimation, on which several of the above methods depend, becomes unreliable in low-contrast or heavily degraded regions, creating a circular dependency in which the filter needed to recover the orientation field itself requires a reasonably clean image.

In combination, these limitations mean that classical techniques are effective for mild degradation but degrade in performance precisely where enhancement is most needed, that is, in severely degraded or partial latent prints.

2.4 Deep Learning Based Enhancement

Learning-based approaches replace hand-crafted filters with models trained on paired or unpaired degraded/clean examples. Three broad families are relevant to this work.

Approach	Mechanism	Typical Limitation
Convolutional Neural Network (CNN)	Learns a direct pixel-wise mapping minimizing MSE or MAE	Regresses toward the mean; outputs are blurred and lose minutiae
Autoencoder	Compresses input to a latent representation and reconstructs it	Bottleneck discards fine, high-frequency ridge detail
Conditional GAN (Pix2Pix)	Learns the mapping under an adversarial realism constraint	Requires paired data and careful loss balancing, but preserves sharp, realistic local structure

A **conventional CNN or autoencoder** trained with a purely pixel-wise reconstruction loss such as **mean squared error tends to average over plausible ridge continuations wherever the input is ambiguous**, producing an output that is smooth and structurally plausible on average but locally blurred. For fingerprints, this is a critical failure mode because minutiae are precisely the small, locally distinctive structures, ridge endings and bifurcations, that such averaging destroys. This motivates the use of an adversarial loss in addition to, rather than instead of, a pixel-wise loss.

2.5 Conditional Generative Adversarial Network Theory

A standard Generative Adversarial Network (GAN), as formulated by Goodfellow et al., consists of two networks trained in opposition. A generator G maps a random noise vector to a data sample, and a discriminator D is trained to distinguish generated samples from real data samples. The two networks are trained as players in a minimax game:

$$\min_G \max_D E_x[\log D(x)] + E_z[\log(1 - D(G(z)))]$$

In this formulation, the generator has no mechanism to control what it produces beyond the random noise input, which is unsuitable for a restoration task where the output must correspond to a specific input image. A Conditional GAN (cGAN) resolves this by conditioning both the generator and the discriminator on an observed input x , in this case the degraded fingerprint, so that the generator learns a mapping from a specific degraded image to a specific restored image rather than an unconstrained data distribution.

$$\min_G \max_D E_{\{x,y\}}[\log D(x,y)] + E_{\{x,z\}}[\log(1 - D(x, G(x,z)))]$$

Here, x is the degraded fingerprint, y is the corresponding clean SOCOFing ground truth, and z represents stochasticity in the generator, in practice implemented through dropout at inference time rather than an explicit noise vector. The discriminator $D(x,y)$ is trained to output a high probability when y is a genuine clean fingerprint corresponding to x , and a low probability when the second argument is a generator output $G(x,z)$. The generator is simultaneously trained to maximize the probability that the discriminator assigns to its outputs, effectively learning to produce restorations that a well-trained discriminator cannot distinguish from genuine clean prints. Over the course of training, the generator improves because the discriminator becomes progressively more discerning, and the discriminator improves because the generator becomes progressively more convincing, a dynamic equilibrium that is the defining characteristic of adversarial training.

2.6 Pix2Pix Architecture

2.6.1 U-Net Generator

The Pix2Pix generator is a U-Net, an encoder-decoder architecture augmented with skip connections between mirrored encoder and decoder layers. The encoder progressively downsamples the input degraded fingerprint through a sequence of convolutional layers, extracting increasingly abstract, spatially coarse feature representations, and compressing the image into a low-dimensional bottleneck. The decoder then progressively upsamples this representation back to the original resolution, reconstructing the enhanced fingerprint.

Degraded Input --> Encoder (Downsampling) --> Bottleneck --> Decoder (Upsampling) --> Enhanced Output

Without skip connections, an encoder-decoder architecture is forced to route all information through the narrow bottleneck, which discards fine spatial detail, precisely the ridge-level and minutia-level detail that this application must preserve. Skip connections address this by directly concatenating the feature maps of each encoder layer with the corresponding decoder layer at the same spatial resolution, allowing low-level information such as ridge position, ridge spacing, and local orientation to bypass the bottleneck entirely. This design is particularly well suited to fingerprint restoration because it preserves ridge continuity and minutiae locations while still allowing the bottleneck to encode the higher-level structural context needed to plausibly complete occluded or heavily degraded regions.

2.6.2 PatchGAN Discriminator

Rather than classifying an entire image as real or fake with a single scalar output, Pix2Pix uses a PatchGAN discriminator, which classifies each local patch of the image independently and averages the resulting predictions. For an image divided conceptually into N by N patches, the discriminator outputs a real/fake decision for every patch rather than for the image as a whole. This design choice is motivated by the observation that high-frequency, local structure, such as ridge texture, is what most strongly signals realism in image restoration tasks, while global, low-frequency structure is already reasonably well captured by the L1 loss described in the next section. By focusing the adversarial signal on local patches, PatchGAN encourages the generator to produce sharp, locally realistic ridge texture throughout the image rather than only in a few salient regions, which is directly relevant to fingerprint restoration, where minutiae can occur anywhere within the print.

2.7 Loss Functions

The Pix2Pix generator is trained under a combined objective that balances two complementary goals: producing outputs that are realistic according to the discriminator, and producing outputs that are faithful to the specific ground-truth identity of the input fingerprint.

2.7.1 Adversarial Loss

The adversarial component of the loss is the conditional GAN objective introduced in Section 2.5, and is responsible for encouraging the generator to produce outputs that are indistinguishable, at the patch level, from genuine clean fingerprints. Without this term, the generator has no incentive to produce sharp, realistic ridge texture, and tends to default toward smooth, low-frequency outputs.

2.7.2 L1 Reconstruction Loss

The reconstruction component penalizes the absolute pixel-wise difference between the generated output and the ground-truth SOCOFing image:

$$L_{L1}(G) = E_{\{x,y,z\}}[||y - G(x,z)||_1]$$

An L1 penalty is preferred over an L2 (mean squared error) penalty because it produces less aggressive smoothing of high-frequency detail, which better preserves the sharp ridge boundaries required for minutiae extraction. This term ensures that the enhanced fingerprint corresponds to the correct underlying identity rather than merely looking like a plausible fingerprint in general.

2.7.3 Combined Objective

The full generator objective combines both terms, with a weighting hyperparameter that balances realism against fidelity:

$$G^* = \arg \min_G \max_D L_{cGAN}(G, D) + \lambda \cdot L_{L1}(G)$$

The weighting hyperparameter λ is conventionally set to a large value, such as 100, so that the L1 term dominates the overall gradient magnitude while the adversarial term continues to sharpen fine texture. This weighting is particularly appropriate for a forensic application, where preserving the true identity of the fingerprint, and therefore favouring fidelity over unconstrained realism, is the primary requirement, and realism serves mainly to avoid the blurring artefacts of a pure L1-only regression model.

2.8 Fingerprint-Specific Advantages of the Pix2Pix Formulation

The architectural and loss design choices described above are collectively well matched to the specific requirements of fingerprint restoration, for the following reasons:

- U-Net skip connections preserve ridge continuity and ridge spacing, which are the geometric properties from which minutiae are ultimately derived.
- PatchGAN discrimination enforces locally realistic texture throughout the image, rather than only in visually dominant regions, which matters because minutiae can be located anywhere within the print.
- The L1 fidelity term anchors the output to the correct fingerprint identity, preventing the network from generating plausible but incorrect ridge patterns in heavily degraded or occluded regions.
- Paired supervision, made possible by the synthetic degradation framework of Section 1, allows the network to learn a direct and identity-consistent degraded-to-clean mapping, rather than an unpaired or style-only translation.

2.9 Expected Influence on YOLO-NAS Based Minutiae Detection

Following enhancement, the restored fingerprint is intended to be passed to a YOLO-NAS based minutiae localization model. Improved ridge clarity is expected to influence this stage through the following causal chain:

Improved Ridge Clarity --> Higher Minutiae Confidence --> More Precise Bounding Box Localization --> Improved Precision and Recall
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Sharper, more continuous ridge structures should reduce the ambiguity around ridge endings and bifurcations, which in turn should increase the confidence scores assigned by the detector to genuine minutiae, reduce the number of missed detections (false negatives) in degraded regions, and reduce spurious detections (false positives) caused by noise being misinterpreted as ridge discontinuities. These effects are expected to jointly improve both precision and recall of the downstream detector, although this report presents them as theoretically motivated expectations to be verified experimentally rather than as measured outcomes.

2.10 Expected Influence on SourceAFIS Based Matching

The enhanced fingerprint and its associated minutiae are ultimately intended to be evaluated using SourceAFIS, an open-source minutiae-based matching engine. The anticipated benefit chain for this stage is as follows:

Cleaner Minutiae Set --> More Stable Descriptors --> Better Point Correspondence --> Higher Genuine Match Scores and Lower False Non-Match Rate

A minutiae set that is less contaminated by spurious or missing points should produce more stable local descriptors, which should in turn improve the reliability of point correspondence between probe and reference templates during matching. The expected net effect is an increase in genuine match scores, a reduction in the False Non-Match Rate (FNMR) for genuinely matching but degraded pairs, and an overall improvement in the Receiver Operating Characteristic (ROC) of the matching system when evaluated on degraded or latent-style inputs relative to an unenhanced baseline.

2.11 Expected Experimental Outcomes

Based on the theoretical properties described above, the following outcomes are anticipated once the Pix2Pix model described in this report is implemented and evaluated, and are presented here as hypotheses to be tested rather than as results:

- Enhanced fingerprints should achieve higher Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) relative to their degraded counterparts, when compared against the SOCOFing ground truth.
- Minutiae detection precision and recall should improve on enhanced images relative to unenhanced degraded images, particularly at the severe degradation level defined in Report 2.
- Matching accuracy under SourceAFIS should improve most noticeably on the severe degradation subset, since this is the regime in which classical enhancement is expected to fail most sharply.
- Improvements are expected to be smaller, though still present, on the mild degradation subset, where classical enhancement already performs reasonably well.

Conclusion:

This report has developed the theoretical foundation required for the next implementation phase of the project. Section 1 refined the synthetic degradation framework established in Report 2 by incorporating PolyU-derived sensor noise and background texture through a spatially varying alpha compositing scheme, reducing the domain gap between synthetically degraded training pairs and genuine forensic latent impressions. Section 2 developed the complete theoretical basis of the Pix2Pix conditional GAN, covering the limitations of classical and naive deep learning enhancement approaches, the minimax formulation of conditional adversarial training, the U-Net generator and PatchGAN discriminator architectures, the combined adversarial and L1 loss objective, and the theoretically expected impact of enhancement on downstream YOLO-NAS minutiae detection and SourceAFIS matching performance.

Together, these two sections provide the methodological basis for implementing, training, and evaluating the Pix2Pix enhancement module against the paired SOCOFing dataset prepared in Report 2. The next phase of this work will implement the described architecture in PyTorch, train it on the refined paired dataset, and evaluate it quantitatively using PSNR, SSIM, and downstream minutiae detection and matching performance, in preparation for the major research paper to follow.

References:

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